

MXEN3005 Robot Manipulation Project
2026 Semester 1

Design Project

ROS2 package for teleoperation of Dobot Magician

Total marks	30
Scheduled time	Weeks 11 – 14
Demonstrations	Week 14 (during your last lab session)
Latest report submission	Study week (Sunday, May 31)
R: Report submission	D: Demonstration in lab

The submission must include the report and zipped directory with commented source code of your package. **Note:** do not zip the entire workspace, just the source directory.

1. Project description

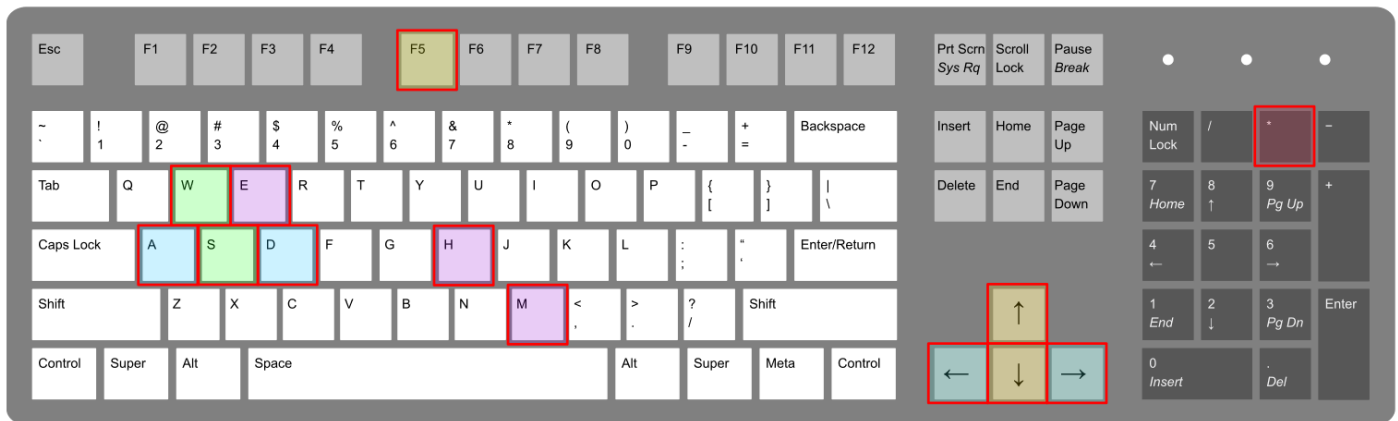
This is a group project where students work on the design and implementation of ROS2 package for teleoperation of Dobot Magician across the lab 205.126 network. Specific requirements for the package are listed below.

- The package name must be “dobot_teleop”.
- The package will be used on two computers: the station you are working on every week (“remote computer”) and the standalone Ubuntu desktop connected to Dobot Magician (“dobot computer”).
- The package must implement two launch files: “dobot_teleop_launch.xml” and “dobot_teleop_keyboard_launch.xml”.
- The launch file “dobot_teleop_launch.xml” will be run on the “dobot computer” and must implement node(s) for sensing and actuation.
- The launch file “dobot_teleop_keyboard_launch.xml” will be run on the “remote computer” and must implement node(s) for teleoperation and visualisation.
- The nodes started in both launch files must communicate with each other using appropriate interface(s) and provide the required functionalities (see section 2).

2. Assessed functionalities (17 marks)

The nodes started in the launch files must provide teleoperation and visualisation functionalities for Dobot Magician connected to “dobot computer” by using keyboard and monitor connected to “remote computer”. To simplify and standardise the use of keyboard on the “remote computer” for all students, a python file “keyboard.py” is provided on Blackboard. The file implements class “Keyboard” that allows for checking if a key

is up or down. Please refer to the Design Project Handout document for more details on how to use this class. Note that the default ROS2 package “teleop_twist_keyboard” can only detect if a key is down (not up) and therefore it is not suitable for fine control of manipulators. The required key mappings to be implemented in this design project are illustrated on the keyboard layout below.



Keypad asterisk * = enable/disable keyboard control

Left/Right arrows = control **joint 1** (joint mode) or **y** (Cartesian mode)

Up/Down arrows = control **joint 2** (joint mode) or **x** (Cartesian mode)

W/S = control **joint 3** (joint mode) or **z** (Cartesian mode)

A/D = control **joint 4** (joint mode) or **r** (Cartesian mode)

E = enable/disable suction cup

H = start homing procedure

M = toggle between joint and Cartesian modes

F5 = perform automated block stacking

The marking scheme for assessed functionalities is listed below.

Functionality	Marks
Correct, robust and user-friendly control in joint mode	4 (1 mark per joint)
Correct, robust and user-friendly control in Cartesian mode	9 (3 marks per x, y, z)
Correct and robust trigger/toggle controls	2 (0.5 mark per trigger/toggle)
Correct visualisation (digital twin)	2

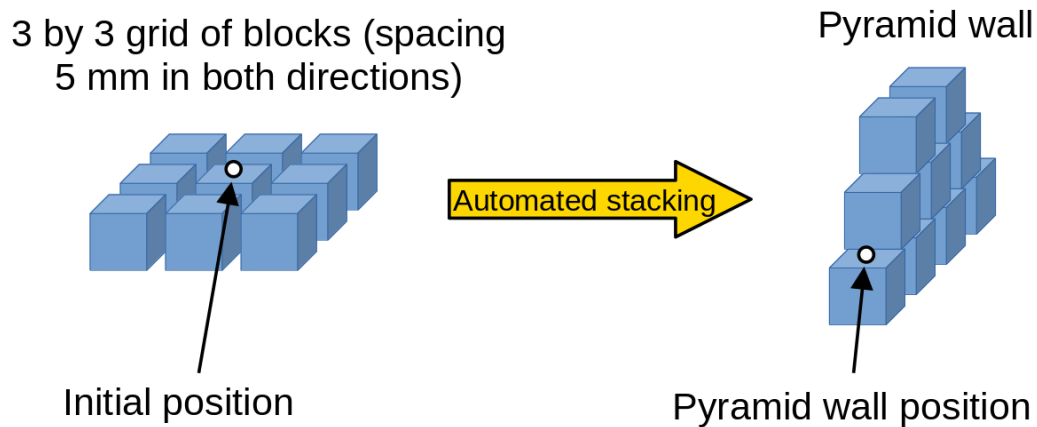
3. Demonstration (8 marks)

Each group must demonstrate teleoperation of Dobot Magician using their package. There will be two demonstrations, both carried out on the “remote computer” and assessed on the “dobot computer”.

Demonstration 1 – automated block stacking (6 marks)

Students will be asked to home the Dobot and then start automated block stacking procedure by pressing “F5” key. The automated block stacking procedure must move 9 blocks arranged in a 3 by 3 grid (spacing 5 mm in both directions) and stack them in a pyramid wall with the rows from bottom to top consisting of 4, 3 and 2

blocks, respectively. The initial position of the centre block in the grid is $x = 92 \text{ mm}$, $y = -201 \text{ mm}$. The position of the first block in the bottom row of the pyramid wall is $x = 135 \text{ mm}$, $y = 226 \text{ mm}$. The automated block stacking procedure is illustrated below.



Demonstration 2 – manual placement of the final block (2 marks)

Students will be asked to manually place one final block on top of the pyramid wall in the mode of their choice (joint or Cartesian).

The marking scheme for both demonstrations is listed below.

Demonstration	Marks
Correct pick positions and safe trajectories to the place positions	3
Correct stacking of pyramid wall	3
Successful placement of the final block	2
Unsuccessful placement of the final block without knocking down pyramid wall	0
Knocking down pyramid wall	-2

4. Report (5 marks)

The report must be a neatly formatted document that is no more than 5 pages long and provides a clear description of the package design and the flowcharts describing how teleoperation has been achieved in joint and Cartesian modes.

The marking scheme for report is listed below.

Report content	Marks
Package design in the form of ROS2 graph (include all nodes and interfaces used)	2
Flowcharts describing how a key press is converted to motion in joint and Cartesian modes	2
Structure, formatting, clarity	1